Negar Haghpanahi

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Research Summary

PhD student and Graduate Research Assistant in Computer Science specializing in **Edge-AI**, **adaptive inference**, and **hardware-efficient ML**. Designs early-exit frameworks (AdaExit, SEE-RF, DIAL) achieving up to **60% energy reduction** with <1% accuracy loss on resource-constrained IoT, wearable, and energy-harvesting devices, and builds **clinical AI**, including a digital twin and prognostic agent for severe traumatic brain injury developed with clinical partners. Experienced in **deep learning**, **LLM systems**, and **model optimization** across research and production.

Education

Washington State University Jan 2025 – Present

*PhD (Direct Path), Computer Science | GPA: **3.98/4.0** | Advisor: Dr. Ganapati Bhat* *Pullman, WA*

- M.S. in Computer Science earned en route; defense in Fall 2026

Amirkabir University of Technology (Top 3 Nationally) Sep 2019 – Jul 2023

*B.Sc. Computer Science | CGPA: **3.55/4.0*** *Tehran, Iran*

- Selected coursework: Numerical Linear Algebra (A+), Linear Optimization (A+), Principles of Computer Systems (A+), Computer Simulation (A+), Principles of Software Design (A+), Data Mining (A), Computer Networks (A)

Publications

Energy-Efficient Time Series Applications on IoT Devices via Sensor-Aware Early Exit Random Forest Architectures (SEE-RF) **ESWeek 2026 (Published)**

Belhaj, S.*, Haghpanahi, N.*, Nelson, and Bhat, G. L., et al.

*Equal contribution

A Digital Twin for Severe Traumatic Brain Injury Prognosis (Poster) **ACRM 2026**

Haghpanahi, N., and Bhat, G. et al.

Adaptive Early Exits with Dynamic Sampling Frequency for Efficient Time-Series Classification **Under Review**

Haghpanahi, N., Alizadehziri, F., and Bhat, G.

Designing for Accelerated Translation of a Prognostic Agent in Traumatic Brain Injury **Final Stages of Preparation**

Haghpanahi, N., Jackson, F., Kemp, A., and Bhat, G.

DIAL: Re-imagining Energy Management in Energy-Harvesting IoT Devices via Early Exit Classifiers **Final Stages of Preparation**

Haghpanahi, N., Alizadehziri, F. and Bhat, G.

Research Experience

Washington State University | Graduate Research Assistant | Pullman, WA Jan 2025 – Present

- Engineered **AdaExit** and **SEE-RF** early-exit architectures, slashing edge device energy consumption by **50–60%** with **<1% accuracy degradation** across 6 wearable-sensor benchmarks
- Developed **DIAL**, re-imagining energy management for **energy-harvesting IoT devices** with early-exit classifiers, and combined early exits with **dynamic sampling frequency** for efficient time-series classification
- Built a **clinical digital twin** and **prognostic agent** for severe TBI in collaboration with St. Luke's Rehabilitation Institute and the WSU Elson S. Floyd College of Medicine
- Optimized on-device model inference latency for real-time health-monitoring systems and streamlined deep learning training cycles on **Kamiak HPC clusters** via efficient resource allocation

Augusta University | Research Collaborator with Dr. R. Rahaeimehr | Remote Oct 2022 – 2024

- **ML for Differential Diagnosis of Ear Disease:** applied machine learning to complex clinical data to support differential diagnosis

- **Acoustic Side-Channel Attack on Keyboards:** contributed to keystroke-inference research and collected **25+ keyboard-recording datasets**

Research Projects

EMI: Energy-Aware Early-Exit for Wearable IoT | *PyTorch, TensorFlow* 2026

- Applied **early-exit inference** with **Bayesian optimization** to derive a **Pareto-optimal** energy–accuracy frontier and a **lookup-table controller** for runtime exit selection, deploying **AdaExit** on **NVIDIA GPU** and **Raspberry Pi** edge hardware

Clinical Digital Twin for Severe TBI Prognosis | *Python, scikit-learn, PyTorch, XGBoost* 2026

- Built a **digital twin** and **prognostic pipeline** on the **TBIMS** registry (∼17K patients) with **missing-data imputation**, ensemble **feature selection**, and 12+ benchmarked **ML models**, applying **reinforcement learning (RL)** for sequential decision policies

AdaExit: Adaptive Early-Exit Inference for Edge Devices | *PyTorch* 2025

- Designed a **confidence-based early-exit** framework for **adaptive on-device inference**, cutting **energy** 50–60% at **<1% accuracy loss** across 6 wearable-sensor benchmarks

LLM-Augmented Next-Action Prediction | *PyTorch, Transformers* Nov 2025

- Benchmarked **LSTM** vs. **LLM-augmented** models on **CASAS** smart-home data, analyzing **edge-latency** vs. accuracy trade-offs with confusion-matrix and Sankey analysis

Industry Experience

BizGPT | AI Developer | Toronto, Canada (Remote) Apr 2024 – Aug 2024

- Architected an end-to-end **LLM-powered chatbot** using **LangChain** and **Hugging Face**, automating complex enterprise conversational workflows to reduce manual routing overhead
- Developed automated ETL pipelines using **Apache Airflow**, increasing production deployment reliability while minimizing API token consumption and model execution latency

Bahar Ertebat Gostar | Data Engineer Sep 2022 – Aug 2024

- Built supervised machine learning and clustering pipelines in Python, driving a **15% increase in predictive accuracy** for core business intelligence and downstream analytics tasks
- Refactored legacy monolithic ML architectures into modular, reusable design patterns, accelerating team development velocity by **35%** and optimizing unstructured data streams via **MongoDB** integration

Bahar Ertebat Gostar | BI Developer Intern Mar 2022 – Aug 2022

- Profiled and optimized 30+ complex **SQL** queries using **SSMS Execution Plans**, delivering a **45% reduction in database execution latency** and query processing time
- Re-engineered relational database schemas with optimized joins, partitioning, and advanced indexing for high-throughput enterprise data pipelines

Teaching Assistant

Washington State University – TA for Big Data, Advanced Data Structures, Cyber Security, Computer Programming, Database Systems, Program Design & Development Jan 2025 – May 2026

Amirkabir University of Technology – TA for Numerical Linear Algebra (35+ students), Principles of Software Design, Programming Languages (C, Python) Sep 2022 – Jul 2024

Awards & Certifications

Winner, Call for Ideas, WA State Academy of Sciences – Energy-neutral edge AI for agriculture Sep 2026

2nd Place Winner, AgAID Digital AgAthon – Developed predictive ML workflows for crop residue coverage Feb 2025

NVIDIA Fundamentals of Deep Learning Certification Apr 2025

Professional Service

Reviewer, *IEEE Design & Test* Oct 2026

Technical Skills

ML & Deep Learning: PyTorch, TensorFlow, PyTorch Geometric, Scikit-learn, XGBoost, SHAP, CNNs, LSTMs, Transformers, early-exit networks, model compression, quantization, Bayesian optimization, reinforcement learning
LLM & Generative AI Systems: LangChain, Hugging Face, Transformers, RAG, Agentic AI, On-Device AI Agents
MLOps & Systems: Python, C/C++, SQL, Linux, Git, Slurm/HPC, Apache Airflow, MongoDB, NVIDIA GPU & Raspberry Pi deployment
Other: C#, R, MATLAB, Power BI, Selenium, BeautifulSoup

Additional Projects

Web Scraping & Markdown Conversion Pipeline | *Selenium, BeautifulSoup* Jul 2024

- Concurrent pipeline converting **10K+ web pages** into structured Markdown for downstream LLM fine-tuning

Breast Cancer Prediction with MLP | *TensorFlow* Jun 2023

- MLP with dropout regularization and hyperparameter tuning achieving **97% test accuracy**

Classical ML & Data Modeling | *scikit-learn* 2023

- Compared SVM, KNN, and decision trees (accuracy, precision, recall, F1); multiclass clustering with missing-value handling and K-fold validation; random forest and boosting models for rent prediction with SQL Server/MongoDB export

AI Search & Game-Playing Agents | *Python, C* 2021 – 2022

- Minimax agents for Tic-Tac-Toe and Checkers; A*, genetic algorithms, and simulated annealing for 8-Puzzle and predator–prey; Naive Bayes spam detection; lexical analyzer for a B-Minor compiler in C